

STRATUM CORE — DATABASE HEALTH & COMPLETENESS REPORT

OpenAlex Relational Integrity, Entity Normalization & Bibliometric Quality Audit

Target Database:	AUV_v2.db	Active Project:	AUV_v2
Database Path:	/home/krishnakumar/StratumProjects/projects/AUV_v2/data/db/AUV_v2.db	Audit Date:	Sep 23, 2026, 12:14 PM
Corpus Scale:	11,307 papers (2003 – 2024)	Linkage Scope:	OpenAlex Relational Tables + Junction Graph

1. Relational & Corpus Overview

Core volume metrics across papers, normalized entities, and contribution linkages

Entity / Table	Record Count	Linkage & Detail	Corpus Significance
Papers (`papers`)	11,307	Years: 2003 – 2024	Primary bibliographic work records
Authors (`authors`)	23,157	15,242 with ORCID (65.8%)	Unique researcher profiles identified
Institutions (`institutions`)	3,078	3,075 with ROR ID (99.9%)	Normalized academic & corporate affiliations
Countries (`countries`)	107	ISO 3166-1 alpha-2 standard	Geographic entity normalizations
Contributions (`contributions`)	53,062	Author-Paper Junction Rows	Granular co-authorship & affiliation links

2. Data Completeness Audit

Evaluation of critical fields against completeness benchmarks (>=80% Optimal, 50-79% Moderate, <50% Sparse)

Table	Field / Indicator	Available Count	Completeness %	Status
Papers	Abstract Text Present	8,354	73.9%	Moderate
Papers	Country Info Assigned	9,728	86.0%	Optimal
Papers	Institution Info Assigned	9,549	84.5%	Optimal
Authors	ORCID Identifier Assigned	15,242	65.8%	Moderate
Institutions	ROR Identifier Assigned	3,075	99.9%	Optimal
Contributions	Rows with Resolved Country	47,589	89.7%	Optimal
Contributions	Rows with Resolved Institution	46,533	87.7%	Optimal
Contributions	Rows with Raw Affiliation String	44,681	84.2%	Optimal

3. Institution & Country Attribution Coverage

Full = every co-author resolved | Partial = some authors missing | Zero = no authors resolved

Attribution Type	Resolution Level	Papers Count	Share %	Coverage Definition
Institution Attribution	Full (Every Author Matched)	8,840	78.2%	All co-authors mapped to institutions
Institution Attribution	Partial (Some Missing)	709	6.3%	At least 1 matched, but >=1 unassigned
Institution Attribution	Zero (No Authors Matched)	1,758	15.5%	No co-authors mapped to institutions

Attribution Type	Resolution Level	Papers Count	Share %	Coverage Definition
Country Attribution	Full (Every Author Matched)	9,487	83.9%	All co-authors mapped to countries
Country Attribution	Partial (Some Missing)	241	2.1%	At least 1 matched, but >=1 unassigned
Country Attribution	Zero (No Authors Matched)	1,579	14.0%	No co-authors mapped to countries

Zero-Institution Recovery Metric	Value	Basis / Proportion	Recovery Potential
Imputable Rows (with raw affiliation text)	687	13.2% of zero-contribs	Recoverable via Regex / LLM / WoS matching
Dead-End Rows (no affiliation text present)	4,519	86.8% of zero-contribs	Requires external cross-referencing
Papers with DOI Handle	769	43.7% of zero-papers	Can be enriched via Crossref metadata
Papers Imputable (>=1 raw text row)	376	21.4% of zero-papers	High likelihood of partial/full recovery

4. Research Quality & Reach Metrics

Field-normalized citation impact, international reach, and open access distribution

Quality & Impact Indicator	Papers Count	Share %	Benchmark / Scope
Open Access (OA) Published	4,706	41.6%	Free public accessibility
International Collaboration	1,682	14.9%	Co-authors spanning >= 2 countries
Core Journal Publication	8,950	79.2%	Indexed in curated core scientific journals
Top 10% Cited Papers (Field-Normalized)	3,314	29.3%	Citation volume in top 10th percentile
Top 1% Cited Papers (Field-Normalized)	506	4.5%	Citation volume in top 1st percentile
Average Citations per Paper	27.4	—	Mean raw citation count per paper
Average Field-Weighted Impact (FWCI)	4.10	—	World average benchmark = 1.00

Open Access Category	Count	Proportion %	Category Definition
CLOSED	6,601	58.4%	Restricted access behind a subscription paywall
GOLD	2,070	18.3%	Published in a fully open-access journal
DIAMOND	812	7.2%	Open access category
HYBRID	696	6.2%	Published in a subscription journal with an open license
GREEN	659	5.8%	Toll-access paper self-archived in an open repository
BRONZE	469	4.1%	Free to read on publisher page without formal license

5. Primary Research Topics Distribution

Top academic topics classified by OpenAlex topic models

Rank	Topic Name	Papers Count	% Share
#1	Underwater Vehicles and Communication Systems	4,843	42.8%
#2	Adaptive Control of Nonlinear Systems	1,358	12.0%

Rank	Topic Name	Papers Count	% Share
#3	Robotic Path Planning Algorithms	775	6.9%
#4	Maritime Navigation and Safety	587	5.2%
#5	Distributed Control Multi-Agent Systems	531	4.7%